

Quadcopter Trajectory Tracking in the Presence of 4 Faulty Actuators: A Nonlinear MHE and MPC Approach

Akram Eltrabyly^{ID}, Dalil Ichalal^{ID}, and Said Mammar^{ID}

Abstract—This letter proposes an Active Fault Tolerant Control (AFTC) framework for a quadcopter Unmanned Aerial Vehicle (UAV). The proposed framework is capable of achieving trajectory tracking, estimating states, as well as estimating and compensating for all four actuators multiplicative faults. The framework combines two optimization-based nonlinear estimation and control techniques; Nonlinear Moving Horizon Estimation (NMHE) and Nonlinear Model Predictive Control (NMPC). We formulate the NMHE algorithm such that it simultaneously estimates the states and the 4 actuators faults. These estimates are then provided to NMPC to achieve fault accommodation. To validate our proposed framework, we investigate two fault scenarios, where 4 actuators experience an abrupt and time-varying partial Loss Of Effectiveness (LOE) in the presence of noisy measurements. Finally, simulation results show satisfactory performance and prove that the proposed framework is numerically tractable and real-time applicable.

Index Terms—Fault estimation, fault-tolerant control, nonlinear model predictive control, nonlinear moving horizon estimation.

I. INTRODUCTION

TODAY, many Unmanned Aerial Vehicles (UAVs) applications, such as drone light shows and sports event filming, have close interactions with humans. Thus, a high risk of injury is expected should the drone have any faults. This increases the demands for the drones' safety and reliability. Such faults can be handled using an Active Fault Tolerant Control (AFTC) framework [1].

An AFTC framework combines a Fault Detection and Diagnosis (FDD) module and a reconfigurable controller module. The latter receives fault estimation from the FDD module to achieve fault accommodation. The time between the occurrence of a fault and its delivery to the reconfigurable controller is critical for system stability. For this reason, It is essential

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The authors are with the Univ Evry, IBISC, Université Paris-Saclay, 91020 Evry-Courcouronnes, France (e-mail: akram.eltrabyly@univ-evry.fr; dalil.ichalal@univ-evry.fr; said.mammar@univ-evry.fr).

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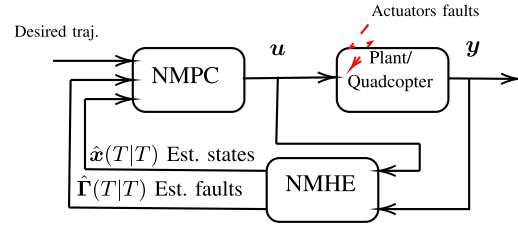


Fig. 1. Active Fault Tolerant Control framework.

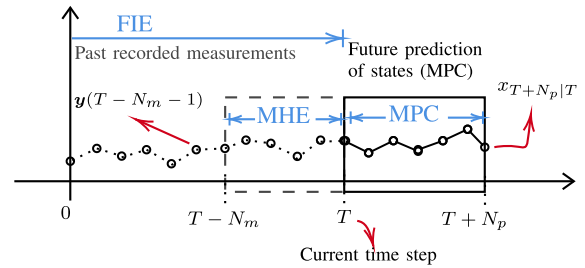


Fig. 2. Horizons of FIE, MHE, and MPC.

to employ an accurate and fast enough FDD module (fault estimator).

Fig. 1 illustrates the components of the presented AFTC framework. We use Nonlinear Moving Horizon Estimation (NMHE) as an FDD module to estimate the states and fault parameters (actuators effectiveness), and feed them into Nonlinear Model Predictive Control (NMPC). The reconfigurable controller module, which is based on NMPC, makes use of the provided estimates to achieve fault accommodation and trajectory tracking for a quadcopter UAV.

It is well known that Kalman Filter (KF) is an optimal estimator for unconstrained linear systems with normally distributed (Gaussian) process and measurement noises. Yet, it cannot handle constrained nonlinear estimation problems. Another technique capable of handling optimal constrained nonlinear problems is called a Full Information Estimator (FIE) [2], [3]. FIE utilizes all the measurements and control actions from the first time step until the current time step in an optimization problem (Fig. 2) to find optimal estimates of states (and potentially parameters). As time proceeds, the optimization problem grows unbounded, turning into an intractable problem. Moving Horizon Estimation (MHE) manages to bound the optimization problem by using a finite

fixed-length horizon (window) of data, backward in time from the current time step (Fig. 2), and discards the rest of the data [4]. However, the discarded data can be approximated using an approximate arrival cost to attract MHE performance as close as possible to FIE [2]. The choice of the approximate arrival cost affects the performance and stability of MHE. The reader is referred to [2], [3], [5] for further details on the choice of the arrival cost.

The superiority of MHE to Extended Kalman Filter (EKF) has been demonstrated in [2] in multiple aspects. For instance, MHE provides greater accuracy because it employs a nonlinear model rather than linearization. In terms of robustness to poor initial guesses, MHE outperforms EKF. Furthermore, MHE distinguishes itself from the majority of the estimation algorithms by its ability to handle constraints and non-gaussian noise distributions. The aforementioned facts motivate our preference for MHE over its competitors.

It is generally admitted that accurate state estimates are important for the success of any feedback controller. Nevertheless, most state estimators exhibit degraded performance in the presence of faults since the estimator was primarily developed for the fault-free model. This alone can cause system instability by having skewed state estimates not taking the post-fault model into consideration. Fortunately, MHE is capable of estimating faults and states simultaneously, thus providing reliable estimates compared to other techniques not taking fault into account.

In the past two decades, several works employed MHE for fault estimation. Reference [6] used MHE for fault detection in hybrid systems. Reference [7] estimates fault parameters of a UAV with MHE. MHE was utilized without an arrival cost in [8] to estimate actuator fault parameters for a quadcopter. However, they removed its state estimation capabilities to reduce the computational cost. Reference [9] presented linear MHE for fault estimation in renewable microgrids. In our work, we implement nonlinear MHE for a quadcopter UAV as an FDD module to estimate the states and all 4 actuators faults simultaneously in the presence of noisy measurements.

Multiple control strategies have been utilized as reconfigurable controllers in the literature. However, MPC poses several unique properties that set it apart from its rivals. Firstly, MPC has the ability to incorporate a nonlinear dynamic model as its internal model, thus offering greater accuracy. Secondly, since MPC solves an optimization problem subject to system dynamics and constraints at each time step, it is able to exploit estimated fault parameters to update its internal model to reflect the post-fault model. This enables MPC to reconfigure the controller online. Thirdly, MPC's ability to enforce state and control input constraints is highly appreciated for fault-tolerant control. This is due to the additional physical limitations faults can induce. For instance, an actuator experiencing a partial loss of effectiveness will suffer a reduction in the amount of force it can generate. To compensate for that loss, control actions may become larger, potentially exceeding the saturation limits. Finally, [10] shows that MPC with control inputs constraints demonstrates a degree of implicit fault-tolerance capability.

MPC and MHE are similar in that both are optimization-based fixed-length moving horizon techniques for control and estimation, respectively. Fig. 2 shows the

relative location of the fixed-length moving horizons of MPC and MHE problems solved at the current time step T . MHE uses a horizon with N_m steps that extend backward from the current instant T . On the other hand, MPC, as a 'predictive' method, uses a horizon of N_p steps that extend forward in the future for prediction. In addition, Fig. 2 demonstrates the difference between the number of measurements utilized in MHE and FIE, where MHE employs a fixed-length moving horizon, and FIE employs an expanding horizon.

Main contributions: a) This letter addresses a gap in the literature by using NMHE to simultaneously estimate the states and 4 actuators multiplicative faults of a quadcopter. b) We integrate NMPC and NMHE to present a complete AFTC framework capable of achieving trajectory tracking in the presence of 4 actuators multiplicative faults and noisy measurements. c) We provide illustrations (Fig. 2, and Fig. 4) and discussions comparing the horizons of FIE, MHE, and MPC and simplifying the understanding of the MHE smoothing scheme.

II. QUADCOPTER MODEL WITH FAULT REPRESENTATION

A. Notations

- Bolded letters designate vectors.
- Matrices are represented by upper-case letters.
- Scalars are depicted by Non-bolded, lower case letters.
- Superscripts B and I denote the body and inertial frames, respectively.

For instance, x is one component of the quadcopter's position, while $\mathbf{x}$ is the state vector.

B. Nonlinear Model

The quadcopter's nonlinear model can be deduced using the Newton-Euler approach as follows:

$$\dot{\boldsymbol{\zeta}} = \mathbf{R}_{IB} \mathbf{V} \quad (1a)$$

$$\dot{u} = -gS_\theta + rv - qw - k_x u/m \quad (1b)$$

$$\dot{v} = gS_\phi C_\theta + pw - ru - k_y v/m \quad (1c)$$

$$\dot{w} = F_{th,z}^B/m + qu - pv + gC_\phi C_\theta - k_z w/m \quad (1d)$$

$$\dot{\boldsymbol{\eta}} = \mathbf{T}_{IB} \boldsymbol{\omega} \quad (1e)$$

$$\dot{p}_{I_{xx}} = \tau_{th,x}^B + (I_{yy} - I_{zz})qr - k_p p \quad (1f)$$

$$\dot{q}_{I_{yy}} = \tau_{th,y}^B + (I_{zz} - I_{xx})pr - k_q q \quad (1g)$$

$$\dot{r}_{I_{zz}} = \tau_{th,z}^B + (I_{xx} - I_{yy})pq - k_r r \quad (1h)$$

$\boldsymbol{\zeta} = [x \ y \ z]^T$ represents the position vector of the quadcopter, and $\boldsymbol{\eta} = [\phi \ \theta \ \psi]^T$ describes its orientation using the Euler angles, with roll, pitch, and yaw angles respectively. $\mathbf{V} = [u \ v \ w]^T$ is the translational velocity vector in the body frame. $\boldsymbol{\omega} = [p \ q \ r]^T$ is the body angular velocities vector. g is the gravitational acceleration, m is the quadcopter's mass, k_x, k_y, k_z are positive scalar translational drag coefficients, and k_p, k_q, k_r are positive rotational drag coefficients. I_{xx}, I_{yy}, I_{zz} are the moments of inertia about x, y, z axes respectively. $F_{th,z}^B$ represents the component of thrust force in z axis, resulting from the actuators, and is further detailed in equation (5) along with the torques due to the same

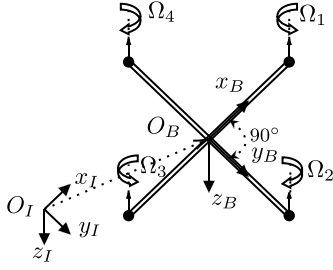


Fig. 3. Rotors' numbers, directions of rotation, and reference frames.

force $\tau_{th,x}^B, \tau_{th,y}^B, \tau_{th,z}^B$.

$$R_{IB} = \begin{bmatrix} C_\theta C_\psi & S_\phi S_\theta C_\psi - C_\phi S_\psi & C_\phi S_\theta C_\psi + S_\phi S_\psi \\ C_\theta S_\psi & S_\phi S_\theta S_\psi + C_\phi C_\psi & C_\phi S_\theta S_\psi - S_\phi C_\psi \\ -S_\theta & S_\phi C_\theta & C_\phi C_\theta \end{bmatrix} \quad (2)$$

$$T_{IB} = \begin{bmatrix} 1 & S_\phi T_\theta & C_\phi T_\theta \\ 0 & C_\phi & -S_\phi \\ 0 & S_\phi / C_\theta & C_\phi / C_\theta \end{bmatrix} \quad (3)$$

R_{IB} and T_{IB} are the rotation and transformation matrices, respectively. $S_\theta = \sin(\theta)$, $C_\theta = \cos(\theta)$, $T_\theta = \tan(\theta)$, and similarly for ϕ, ψ .

C. Multiplicative Fault Representation

To allow for multiplicative fault representation in the aforementioned model, we modify equations (1d), (1f)-(1h) and introduce a fault matrix L [11]:

$$\begin{bmatrix} \dot{w} \\ \dot{p} \\ \dot{q} \\ \dot{r} \end{bmatrix} = M \mathbf{u} + \begin{bmatrix} m(qu - pv) + mgC_\phi C_\theta - k_z w \\ (I_{yy} - I_{zz})qr - k_p p \\ (I_{zz} - I_{xx})pr - k_q q \\ (I_{xx} - I_{yy})pq - k_r r \end{bmatrix} \quad (4)$$

$$\begin{bmatrix} F_{th,z}^B \\ \tau_{th,x}^B \\ \tau_{th,y}^B \\ \tau_{th,z}^B \end{bmatrix} = \underbrace{\begin{bmatrix} -k_T & -k_T & -k_T & -k_T \\ 0 & -k_T l & 0 & k_T l \\ k_T l & 0 & -k_T l & 0 \\ -b & b & -b & b \end{bmatrix}}_M \underbrace{\begin{bmatrix} \Omega_1^2 \\ \Omega_2^2 \\ \Omega_3^2 \\ \Omega_4^2 \end{bmatrix}}_u L \quad (5)$$

M is the allocation matrix that maps the control actions vector $\mathbf{u}$ to the thrust force and torques in the body frame. $\mathbf{u} \in \mathbb{R}^{n_u}$ is the control action vector in (revolutions per minute)², with n_u being the number of control inputs. k_T is the thrust constant, b is the drag constant, and l is the quadcopter's arm length. Fig. 3 shows the rotors' numbers and directions of rotations, along with the body and inertial reference frames.

The diagonal matrix $L = \text{diag}(l_1, l_2, l_3, l_4)$ denotes the actuators effectiveness (Fault Parameters). We define the vector $\mathbf{\Gamma} := [l_1 \ l_2 \ l_3 \ l_4]^T$ such that the diagonal matrix $L := \text{diag}(\mathbf{\Gamma})$. The effectiveness of the i th actuator is therefore $0 \leq l_i \leq 1$, ($i = 1, \dots, 4$), where $l_i = 0$ depicts a complete actuator failure, $l_i = 1$ a healthy actuator, and $0 < l_i < 1$ a partial actuator failure (Loss of Effectiveness). From equations (1), (4), and (5), the nonlinear model with faults and measurement noise is represented as follows:

$$\begin{cases} \dot{\mathbf{x}} = f(\mathbf{x}, \mathbf{u}, L) \end{cases} \quad (6a)$$

$$\begin{cases} \mathbf{y} = h(\mathbf{x}) + \mathbf{v} \end{cases} \quad (6b)$$

With $\mathbf{x} = [\xi^T \ V^T \ \eta^T \ \omega^T]^T \in \mathbb{R}^{n_x}$ the state vector, and n_x the number of states. $\mathbf{y} \in \mathbb{R}^{n_y}$, with n_y the number of

outputs. $\mathbf{v} \in \mathbb{R}^{n_y}$ is random measurement noise. f and h are nonlinear functions.

III. NONLINEAR MOVING HORIZON ESTIMATION

Moving Horizon Estimation (MHE), being an online optimization-based estimator for constrained nonlinear problems, is distinctively useful for the case of fault parameter estimation of multicopters with highly nonlinear dynamics. At each time step, it solves an optimization problem online by minimizing a quadratic cost function over a moving, fixed-size horizon (Fig. 2), to estimate states and parameters over the horizon [6], [12].

We formulate NMHE to simultaneously estimate states and fault parameters. The decision variables of the actuators effectiveness (fault parameters) and the states are represented by the vectors $\mathbf{\Gamma}$ and $\mathbf{X}$, respectively.

We choose the following cost function for NMHE:

$$\begin{aligned} J_T = J_{T-N_m} &+ \sum_{k=T-N_m}^T \left\| \mathbf{y}(k) - h(\mathbf{X}(k)) \right\|_E^2 \\ &+ \sum_{k=T-N_m}^{T-1} \left\| \mathbf{X}(k+1) - f_d(\mathbf{X}(k), \mathbf{u}(k), L(k)) \right\|_{S_1}^2 \\ &+ \sum_{k=T-N_m}^{T-1} \left\| \mathbf{\Gamma}(k+1) - \mathbf{\Gamma}(k) \right\|_{S_2}^2. \end{aligned} \quad (7)$$

The first term, J_{T-N_m} , is called the approximate arrival cost. It approximately accounts for discarded measurements from the first time step until the step before the MHE horizon ($T - N_m - 1$) (Fig. 2). The arrival cost term is to be discussed in more details after introducing MHE optimization problem. The remaining terms penalize measurement noise, process noise, and fault variance, respectively. We define $\|\mathbf{X}\|_{S_1}^2 = \mathbf{X}^T S_1 \mathbf{X}$. The tuning parameters E, S_1, S_2 are typically chosen as the inverse of the noise covariance matrices.

Hence, the problem to be solved at each time step T is:

$$(\hat{\mathbf{x}}(T - N_m : T|T), \hat{\mathbf{\Gamma}}(T - N_m : T|T)) = \arg \min_{\substack{\mathbf{X}(T-N_m : T), \\ \mathbf{\Gamma}(T-N_m : T)}} J_T \quad (8)$$

subject to:

$$\begin{cases} \mathbf{X}(k+1) = f_d(\mathbf{X}(k), \mathbf{u}(k), L(k)) + \mathbf{w}_1(k) \end{cases} \quad (9a)$$

$$\begin{cases} \mathbf{\Gamma}(k+1) = \mathbf{\Gamma}(k) + \mathbf{w}_2(k) \end{cases} \quad (9b)$$

$$\begin{cases} \mathbf{y}(k) = h(\mathbf{X}(k)) + \mathbf{v}(k) \end{cases} \quad (9c)$$

$$\begin{cases} \mathbf{X}_{min} \leq \mathbf{X}(k) \leq \mathbf{X}_{max} \end{cases} \quad (9d)$$

$$\begin{cases} \mathbf{\Gamma}_{min} \leq \mathbf{\Gamma}(k) \leq \mathbf{\Gamma}_{max} \end{cases} \quad (9e)$$

T is the current time step, and N_m is the length of the estimation horizon. $k = T - N_m, \dots, T - 1$ for equations (9a), (9b), and $k = T - N_m, \dots, T$ for equations (9c)-(9e). f_d is determined from equation (6a) by means of numerical integration. $\mathbf{w}_1 \sim \mathcal{N}(0, W_1)$, $\mathbf{w}_2 \sim \mathcal{N}(0, W_2)$, $\mathbf{v} \sim \mathcal{N}(0, V)$ are normal distributions representing process noise, parameters noise, and measurement noise, respectively, with covariance matrices $W_1 \in \mathbb{R}^{n_x}$, $W_2 \in \mathbb{R}^{n_u}$, $V \in \mathbb{R}^{n_y}$. It is worth noting that even when faults are assumed piecewise-constant, including fault covariance ($\mathbf{w}_2$) in equation (9b) allows for the estimation of time-varying faults [13].

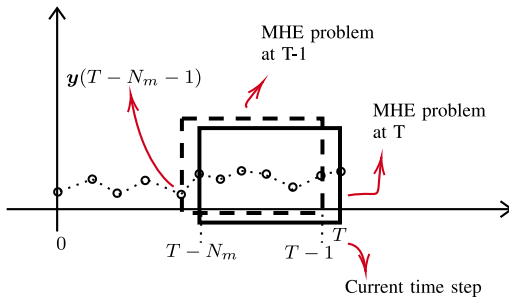


Fig. 4. MHE Smoothing Scheme.

Solving the aforementioned optimization problem, NMHE generates a sequence of estimates of states and faults over the backward horizon, as seen in equation (8). With $\hat{\mathbf{x}}(T - N_m : T|T) = \{\hat{\mathbf{x}}(T - i|T) \mid i = 0, 1, \dots, N_m\}$ the estimated states at each time step over the horizon, and $\hat{\mathbf{\Gamma}}(T - N_m : T|T) = \{\hat{\mathbf{\Gamma}}(T - i|T) \mid i = 0, 1, \dots, N_m\}$ the estimated fault parameters. $\mathbf{X}(T - N_m : T) = \{\mathbf{X}(T - i) \mid i = 0, 1, \dots, N_m\}$, and $\mathbf{\Gamma}(T - N_m : T) = \{\mathbf{\Gamma}(T - i) \mid i = 0, 1, \dots, N_m\}$ are the decision variables of states and fault parameters, respectively, over the backward horizon.

The so-called approximate arrival cost term, J_{T-N_m} , in equation (7) plays a crucial rule in the stability and performance of MHE.

$$J_{T-N_m} = \left\| \begin{matrix} \boldsymbol{\chi}(T - N_m) - \bar{\boldsymbol{x}}(T - N_m) \\ \boldsymbol{\Gamma}(T - N_m) - \bar{\boldsymbol{\Gamma}}(T - N_m) \end{matrix} \right\|_{P^{-1}}^2. \quad (10)$$

To approximate the arrival cost, we choose a smoothing scheme [2], where $\tilde{\mathbf{x}}(T-N_m) = \hat{\mathbf{x}}(T-N_m|T-1)$, $\tilde{\mathbf{\Gamma}}(T-N_m) = \hat{\mathbf{\Gamma}}(T-N_m|T-1)$ are the estimated values of states and fault parameters, respectively, at $T-N_m$ computed at the previous MHE problem at $T-1$. Fig. 4 provides an illustration of the aforementioned smoothing scheme.

Furthermore, authors of [5], [14] discussed multiple ways to calculate the covariance matrix P for arrival cost computation. It is worth noting that the choice of the arrival cost directly affects the performance and stability of MHE. Further details concerning the stability of MHE and its relation to the arrival cost can be found in [2], [3], [5].

IV. FAULT TOLERANT MODEL PREDICTIVE CONTROL

A. Notations

- $\mathbf{u}'_{T+i|T}$, and $\mathbf{x}'_{T+i|T}$ are the decision variables of the control actions and states, respectively, at instant $T+i$ of the MPC problem at instant T .
- $\mathbf{u}_{T+i|T}$, $\mathbf{x}_{T+i|T}$ are the control actions and predicted states at instant $T+i$ found by solving the MPC problem at T (Fig. 2).

B. Fault-Tolerant MPC Problem

The NMPC Optimal Control Problem (OCP) is as follows:

$$\begin{aligned} \min_{U, X} \quad & J(\hat{\mathbf{x}}(T|T), U, \mathbf{X}^{ref}) \\ = \quad & \sum_{i=0}^{N_p-1} \left(\|\tilde{\mathbf{x}}_{T+i|T}\|_Q^2 + \|\mathbf{u}'_{T+i|T}\|_R^2 + \|\Delta \mathbf{u}'_{T+i|T}\|_{R_\Delta}^2 \right) \\ & + \|\tilde{\mathbf{x}}_{T+N_p|T}\|_{Q_N}^2 \end{aligned} \quad (11)$$

TABLE I
PARAMETERS

Parameters	Values	Parameters	Values
$I_{xx} = I_{yy}$	0.0196 kg.m ²	l	0.21 m
I_{zz}	0.0264 kg.m ²	k_r	9e - 3 N.m.s/rad
m	1.1 kg	g	9.81 m/s ²
k_T	9.29e - 5 N.s ²	N_p, N_m	30, 16
b	1.1e - 6 N.m.s ²	$\sqrt{u_{max}}$	18833 rpm

subject to:

$$\left[\begin{array}{l} \mathbf{x}'_{T+i+1|T} = f_d(\mathbf{x}'_{T+i|T}, \mathbf{u}'_{T+i|T}, \hat{L}(T)) \end{array} \right. \quad (12a)$$

$$\left\{ \begin{array}{l} \mathbf{x}_{min} \leq \boldsymbol{\chi}'_{T+i|T} \leq \mathbf{x}_{max} \end{array} \right. \quad (12b)$$

$$\mathbf{u}_{min} \leq \mathbf{u}'_{T+i|T} \leq \mathbf{u}_{max} \quad (12c)$$

$$\text{b} \quad \chi'_{T|T} = \hat{\mathbf{x}}(T|T) \quad (12d)$$

for $i = 0, 1, \dots, N_p - 1$.

NMPC receives the estimated states $\hat{\mathbf{x}}$, and the estimated fault parameters $\hat{\mathbf{L}} := \text{diag}(\hat{\mathbf{\Gamma}})$ from NMHE as demonstrated in Fig. 1. In addition, it receives the reference (desired) state trajectory $\mathbf{X}^{\text{ref}} = \{\mathbf{x}_{T+i}^{\text{ref}} | i = 0, 1, \dots, N_p\}$. NMPC then solves online, at each time step, a finite horizon optimal control problem. The solution returns a sequence of control actions over the future horizon $\mathbf{U} = \{\mathbf{u}_{T+i|T} | i = 0, 1, \dots, N_p - 1\}$, and the predicted state trajectory $\mathbf{X} = \{\mathbf{x}_{T+i|T} | i = 0, 1, \dots, N_p\}$. Only the first step $\mathbf{u}_{T|T}$ of the control actions sequence is implemented in the plant. N_p is the prediction horizon.

$$\begin{aligned}\tilde{\mathbf{x}}_{T+i|T} &= \mathbf{x}'_{T+i|T} - \mathbf{x}_{T+i}^{ref} \\ \Delta \mathbf{u}'_{T+i|T} &= \mathbf{u}'_{T+i|T} - \mathbf{u}'_{T+i-1|T}.\end{aligned}$$

At each instant T at the start of the receding horizon iteration, $\mathbf{X}'_{T|T}$ and $\mathbf{u}'_{T-1|T}$ are known. $\mathbf{X}'_{T|T}$ is set as the states estimated by NMHE $\hat{\mathbf{x}}(T|T)$, and $\mathbf{u}'_{T-1|T}$ is set as the control action from the previous iteration $\mathbf{u}_{T-1|T-1}$. $\mathbf{Q}, \mathbf{Q}_N \in \mathbb{R}^{n_x \times n_x}$, $\mathbf{R}$ and $\mathbf{R}_\Delta \in \mathbb{R}^{n_u \times n_u}$ are diagonal positive definite weighting matrices.

The last term in equation (11) denotes the so-called terminal cost, which is crucial for MPC stability. Authors of [15] demonstrated that if a sufficiently large prediction horizon is employed, applying MPC with a terminal cost is sufficient to ensure stability for nonlinear models without imposing terminal constraints.

In this letter, trajectory tracking is accomplished by providing MPC with the future N_p way-points of the desired trajectory at each time step, as shown in Fig. 1. This enables MPC to use its model-based prediction capabilities to determine appropriate control actions to attain the desired trajectory.

V. RESULTS

In this section, we show the effectiveness of the proposed Active Fault Tolerant Control framework consisting of the two aforementioned optimization-based algorithms, NMHE and NMPC. The AFTC framework is implemented for a quadcopter UAV's nonlinear dynamic system in MATLAB Simulink, along with CasADi framework for nonlinear optimization [16]. [Table I](#) shows the quadrotor's parameters from [17], along with other parameters native to this letter.

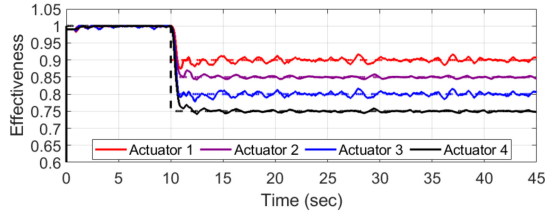


Fig. 5. FS1: estimated actuators effectiveness in solid lines, actual injected effectiveness in dashed lines.

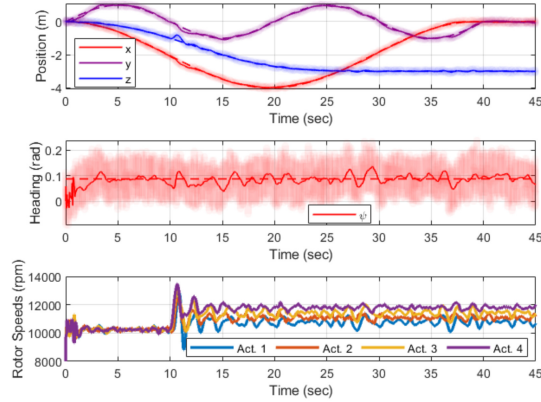


Fig. 6. FS1: (Top) Position, (Middle) Heading, (Bottom) Control Actions. [dashed lines: desired states, transparent: noisy measurements, solid: estimated by NMHE].

Assuming equal measurement noise for all 12 measurements, the measurement standard deviation (SD) is $\sigma_m = 0.0316$. In addition, the process and fault SDs are $\sigma_p = 0.001$, and $\sigma_f = 0.001$, respectively. Using the aforementioned standard deviations, the tuning parameters of NMHE are typically chosen as the inverse of the covariance matrices.

With the goal of increased system safety and reliability, we investigate two tough fault scenarios (FS) as means of testing the proposed framework:

- 1) FS1: An abrupt loss of effectiveness at time $t = 10$ sec occurring to 4 actuators simultaneously, as in Fig. 5.
- 2) FS2: An abrupt loss of effectiveness at time $t = 10$ sec followed by a time-varying fault as in Fig. 8.

A. Fault Scenario 1

In the first fault scenario, all four actuators experience an abrupt loss of effectiveness at the same time, as demonstrated in Fig. 5. Where at time $t = 10$ sec, a fault is injected in all actuators simultaneously, reducing the effectiveness of actuators 1, 2, 3, and 4 to 0.9, 0.85, 0.8, 0.75, respectively.

Fig. 6 (Top, Middle) shows a selected subset of the states: x , y , z (position), and ψ (heading). It shows how the proposed framework could maintain trajectory following with slight deviations around the fault time. The observed deviations are small due to the quick incorporation of fault information into the MPC internal model. It is evident that NMHE succeeded in estimating denoised states from noisy sensor measurements, even in the presence of 4 actuators faults. In addition, It managed to estimate 4 actuators faults simultaneously, as shown in Fig. 5. NMHE has an advantage over its competitors in that it can include constraints on its estimates. For instance, when it comes to actuators effectiveness estimation, other estimators may result in values that are negative or greater than 1.

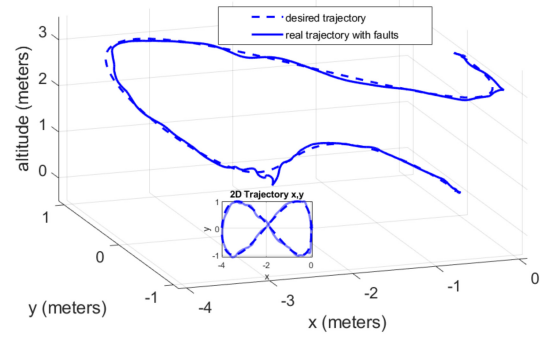


Fig. 7. FS1: 2D and 3D Trajectories.

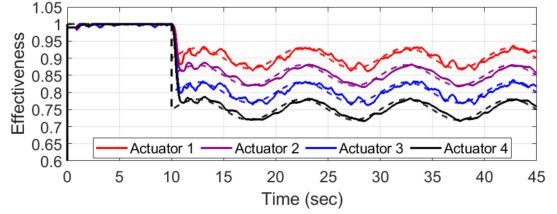


Fig. 8. FS2: estimated actuators effectiveness in solid lines, actual injected effectiveness in dashed lines.

However, NMHE is capable of constraining fault parameters (actuators effectiveness) to be within their practical limit of 0 (complete failure) and 1 (completely healthy).

After receiving the desired trajectories, as well as the aforementioned states and faults estimated by NMHE, NMPC manages to calculate rotor speeds as shown in Fig. 6 (Bottom), to mitigate the effect of faults and follow the desired trajectory. Furthermore, NMPC ensures that control actions do not exceed the actuators' physical limits, even when faults are present. A sudden change in the control actions may be noticed around fault time ($t = 10$ sec) due to the abrupt faults occurring to the system.

A 'lying eight' shaped trajectory tracking in 2D (dimensions) and 3D is depicted in Fig. 7. An acceptable performance is shown, with minimal deviation from the desired trajectory in the presence of measurements noise and abrupt loss of effectiveness faults.

B. Fault Scenario 2

In the second fault scenario (FS), an abrupt LOE at time $t = 10$ sec followed by time-varying faults is injected into the actuators as seen in Fig. 8, where NMHE manages to estimate the states and the time-varying faults in the presence of measurements noise. Once NMPC receives the estimated faults, estimated states, and desired trajectories, it redistributes rotor speeds (see Fig. 9) to minimize the effect of the fault on the system and achieve trajectory tracking. Because NMPC still succeeded in compensating for the time-varying faults, Fig. 9 and Fig. 10 of the states and the 3D trajectory in FS2 turned out similar to their correspondents in FS1.

Table II compares the fault estimation performance of NMHE in fault scenario (FS) 1 and 2 using Root-Mean-Squared Error (RMSE). RMSE is an error indicator, thus lower values of RMSE depict better performance. Because of the time-varying nature of faults in FS2, each actuator in FS2 has a slightly higher RMSE

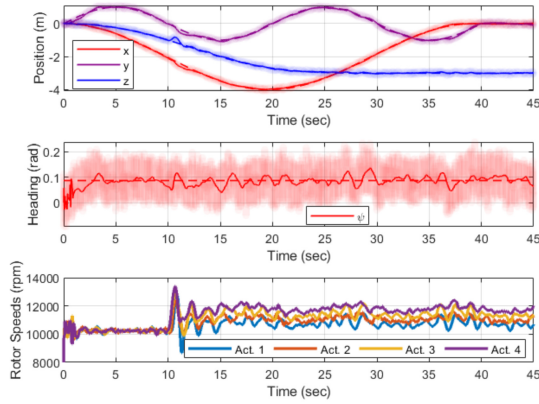


Fig. 9. FS2: (Top) Position, (Middle) Heading, (Bottom) Control Actions. [dashed lines: desired states, transparent: noisy measurements, solid: estimated by NMHE].

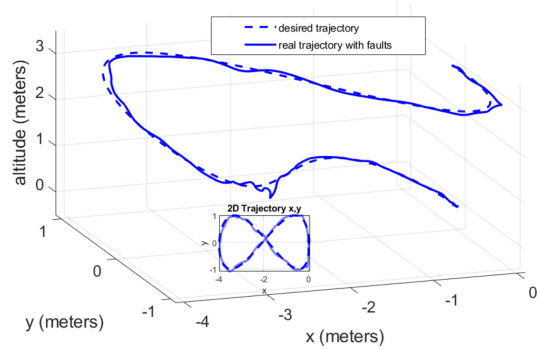


Fig. 10. FS2: 2D and 3D Trajectories.

TABLE II
RMSE OF FAULT AND TRAJECTORY TRACKING-AVERAGE
OF 100 SIMULATIONS

	l_1	l_2	l_3	l_4	Position	ψ
FS1	0.011	0.0136	0.0188	0.022	0.049	0.0199
FS2	0.0129	0.0146	0.0196	0.0225	0.05	0.0199

value than its counterpart in FS1. Moreover, in both fault scenarios, it can be noticed that actuator 4 with the largest loss of effectiveness has the highest RMSE compared to the other three actuators. In addition, Table II shows that the trajectory tracking performance (position and heading) is comparable in both fault scenarios, thus proving the efficacy of the proposed AFTC framework even in the presence of time-varying faults.

The complete framework, including NMHE, NMPC, and the nonlinear quadcopter plant, is real-time implementable in simulation, since it achieves a run-time to simulation-time ratio of 0.71 (less than one). The overall simulation is run with a step-size of 1 ms, while NMPC and NMHE are run at a 50 ms sampling time, on a PC with a 2.6 GHz Intel Core i7-10750H processor and a 16GB memory (RAM).

VI. CONCLUSION

This letter proposed a complete AFTC framework consisting of NMHE and NMPC. NMHE served as an FDD module, estimating the states and all 4 actuators faults of the

UAV, while NMPC represented a reconfigurable controller. NMPC exploited the fault and state estimates provided by NMHE to calculate stabilizing control actions and achieve trajectory tracking. The proposed framework was validated by introducing two challenging fault scenarios that subjected the 4 actuators of the UAV to abrupt Loss Of Effectiveness and time-varying faults. Additionally, measurements were corrupted with random noise. Despite the difficulties introduced, NMHE managed to provide accurate estimations of the states and fault parameters. Simulation results also depicted that NMPC achieved an acceptable trajectory tracking performance and fault compensation. Finally, the two optimization-based nonlinear estimation and control problems were solved online in a real-time fashion on the aforementioned PC.

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